

Basic AI Concepts and Terminologies

A Beginner's Technical Reference Guide

Core definitions and distinctions for artificial intelligence, machine learning, and deep learning — including neural networks, LLMs, inference types, data types, and ML methods. A foundational reference for the AI, ML, and deep learning student.

1. AI, Machine Learning, and Deep Learning

The three terms are often used interchangeably, but they nest inside one another. **Artificial intelligence (AI)** is the overarching field; **machine learning (ML)** is a specialized branch within AI; and **deep learning (DL)** is a further specialization within ML. Getting this hierarchy right is the foundation for everything that follows.

TIP The nesting rule

Think of three concentric circles: **AI** $\supset$ **ML** $\supset$ **DL**. AI is the largest, containing category; deep learning is the innermost. Any statement that makes AI a *subset* or *specialization* of ML is reversed and therefore false.

1.1 Definitions

Term	Definition	Examples
Artificial Intelligence (AI)	Technology that simulates human intelligence to solve problems — making predictions, recognizing images, and generating text.	NLP, computer vision, speech recognition, robotics, expert systems, predictive analytics, automation
Machine Learning (ML)	A type of AI that performs data-analysis tasks without explicit instructions, learning patterns from historical data and improving over time.	Spam detection, image recognition, predictive analytics
Deep Learning (DL)	An ML method that teaches computers to process data in a way inspired by the human brain, using multi-layer neural networks to model complex patterns.	Image & speech recognition, NLP, autonomous driving

1.2 Key differences at a glance

Aspect	AI	ML	Deep Learning
Scope	Overarching field	Subset of AI	Subset of ML
Core idea	Simulate human intelligence	Learn from data	Neural networks on vast data
Programmed explicitly?	Varies	No — learns from data	No — learns from data

Aspect	AI	ML	Deep Learning
Typical use	Broad range of tasks	Predictions & decisions	High-accuracy pattern tasks

KEY POINT 01 The hierarchy runs one way only — AI contains ML contains DL.

A frequent mix-up reverses this hierarchy. ML is a subset of AI, and it is *deep learning* — not AI — that is defined by neural networks. Stated correctly: **deep learning is a subset of ML that uses neural networks to model complex data patterns**, ML is a subset of AI, and AI is the overarching field. Any description that makes AI a subset or specialization of ML is simply reversed.

2. Neural Networks and Core AI Terms

A **neural network** is a deep-learning technique that processes data using interconnected nodes (neurons) arranged in layers, loosely modeled on the human brain. The network learns by adjusting the **weights** on the connections between neurons during training, correcting its mistakes through a process called **backward propagation**.

2.1 The three layer types

Layer	Role
Input layer	Receives the initial data; each neuron represents one feature of the dataset.
Hidden layer(s)	Perform intermediate computations and transformations to identify patterns. More hidden layers = a deeper network (hence “deep learning”).
Output layer	Produces the final predictions or classifications; neuron count usually matches the number of output classes or values.

2.2 Essential terminology

Term	Definition
AI model	A mathematical framework or algorithm that learns patterns from data to perform a task; ranges from simple linear regression to complex neural networks.
AI algorithm	A set of rules or instructions given to an AI model to help it learn on its own.
Computer vision	The ability of computers to extract information and insights from images and videos (e.g. content moderation, facial recognition, image classification).
Natural language processing (NLP)	An ML technology that lets computers interpret, manipulate, and comprehend human language (e.g. sentiment analysis, translation, chatbots).
Training	Teaching a model to make accurate predictions by feeding it large amounts of data so it learns patterns and generalizes to new, unseen data.
Inferencing	Using a trained model to make predictions or decisions on new, unseen data.

Term	Definition
Bias	Systematic errors in a model that produce unfair or inaccurate outcomes.
Fairness	Ensuring a model operates without bias, giving equitable outcomes regardless of protected characteristics.
Fit	The process of training a model on a dataset so it learns the patterns and relationships within it.

WARNING Bias is not just a statistics term

When studying fairness, note that **bias** here refers to systematic errors that cause *unfair or inaccurate* outcomes — a fairness and ethics concern, not only the bias–variance tradeoff. Pair it with **fairness**, its corrective counterpart.

3. Large Language Models (LLMs)

Large language models are very large deep-learning models **pre-trained on vast amounts of data**. They use a **transformer architecture** consisting of an **encoder and a decoder with self-attention capabilities**, which lets them understand and generate human-like text by recognizing patterns and relationships in the data.

LLMs are highly flexible and capable of **unsupervised training** — learning from data without explicit instructions. A single foundation model can be adapted to many tasks.

3.1 What LLMs can do

Input data types	Adapted tasks
Text, images, speech, structured data (databases, spreadsheets), 3D signals	Question answering, sentiment analysis, information extraction, image captioning, object recognition, instruction following, summarization, translation, sentence completion

TIP How to recognize an LLM

When a description stacks these attributes together — *pre-trained + vast data + transformer + encoder/decoder + self-attention* — it is describing a **large language model** specifically, not the broader fields of “NLP” or “computer vision” (those name areas of study, not this particular model type).

4. Inference Types

ML inference is the process of using a trained model to make predictions on new, unseen data. Inference is commonly delivered in four ways; knowing which fits a given scenario is a core practical skill.

Inference type	How it works	Best for
Real-time	Delivers immediate predictions with low latency, one data point at a time.	Instant-response apps: chatbots, recommendation systems, fraud detection , voice assistants

Inference type	How it works	Best for
Serverless	Fully managed infrastructure that auto-scales with traffic; no manual scaling.	Intermittent traffic: mobile apps, web services
Asynchronous	Queues requests and processes them asynchronously; handles large payloads and long run times.	Large-payload / long jobs: video analysis, large-scale data processing
Batch (offline)	Processes large datasets all at once, generating predictions for the whole set.	Offline tasks: nightly recommendations, data preprocessing, business analytics

KEY POINT 02 Real-time and serverless describe different things.

Real-time describes *timing* (instant, low-latency responses). **Serverless** describes *how compute is provisioned* (auto-scaling, no servers to manage). A serverless endpoint can serve real-time requests — but when the requirement is “instant responses,” the relevant concept is **real-time**.

5. Data Types in AI Models

AI uses many data types, each suited to different tasks. The choice depends on the problem and the data available.

Data type	Description	Typical use / examples
Labeled	Includes input features and output labels.	Supervised learning: spam detection, image classification
Unlabeled	Input features but no output labels.	Unsupervised learning: clustering, anomaly detection; raw text
Tabular	Organized in rows and columns (spreadsheet-like).	Classification, regression, clustering; financial & customer records
Time-series	Data points recorded at set time intervals.	Forecasting, anomaly & trend analysis; stock prices, weather, sensors
Image	Images represented as pixel values.	Computer vision: classification, detection, segmentation
Text	Written language.	NLP: sentiment analysis, text classification, translation
Structured	Highly organized and easily searchable; stored in relational, non-relational, or graph databases.	Classification, regression, clustering; relational databases, spreadsheets
Unstructured	Lacks a predefined structure; harder to process.	NLP & computer vision; documents, images, audio, video

TIP Structured vs. tabular vs. unstructured

Tabular is a specific rows-and-columns form of **structured** data. **Unstructured** data (text documents, images, audio, video) has no predefined format. Structured, unstructured, tabular, labeled, unlabeled, time-series, image, and text are all genuine data types — whereas *batch* is a processing mode, not a data type, a distinction worth keeping clear.

6. Machine Learning Methods

There are three main ML methods, distinguished by what kind of data they use and how the model learns.

Method	How it learns	Good for	Advantage / Challenge
Supervised	Trains on a labeled dataset (input paired with correct output).	Classification (spam), regression (house prices)	+ High accuracy on large, representative data. – Needs lots of labeled data (costly).
Unsupervised	Trains on an unlabeled dataset; finds patterns on its own.	Clustering (customer segmentation), anomaly detection (fraud)	+ Works with unlabeled data. – Less interpretable; needs tuning.
Reinforcement	Learns a sequence of decisions via rewards and penalties .	Decision-making over time: game playing (AlphaGo), robotics	+ Learns complex, adaptive behavior. – Compute-heavy; needs a good reward system.

7. Quick Reference: Identifying the Right Concept

Use this quick decision aid to identify which concept a description points to. Each cue on the left maps to the concept it signals on the right — a fast way to keep the categories straight while learning.

- **A statement about the hierarchy of the three terms**
AI is the largest field; ML sits inside AI; DL sits inside ML. The relationship only runs this way — DL is never larger than AI.
- **A description mentioning images or video**
Points to the *field* of **computer vision**. (Deep learning and neural networks are the general *methods* that power it, not the domain itself.)
- **A description mentioning human language (text or speech)**
Points to **natural language processing (NLP)**.
- **A model that is pre-trained on vast data, with a transformer encoder/decoder and self-attention**
Describes a **large language model (LLM)**.
- **A workload needing instant, low-latency responses**
Calls for **real-time** inference — distinct from serverless, which describes how compute scales.

- **A large dataset processed all at once, offline**
Calls for **batch** inference.
- **Data that is labeled vs. unlabeled vs. reward-driven**
Labeled → supervised learning. **Unlabeled** → unsupervised learning. **Rewards / penalties** → reinforcement learning.
- **A term described as a data type**
Structured, unstructured, tabular, labeled, unlabeled, time-series, image, and text are all data types. *Batch* is a processing mode, not a data type.

WARNING Fields vs. methods vs. models

It is easy to confuse terms by placing them in the wrong *category*. **Computer vision** and **NLP** are fields (defined by their data). **Deep learning** and **neural networks** are general methods. **LLMs** and **expert systems** are specific solution types. Keeping these categories distinct is the key to using the terms correctly.

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